

Synchronized visual documentation — four-object semantic-topology experiment

A Outside camera

robot + table + operator
physical outcome and safety context

UTC + monotonic timestamp

B Robot POV

registered RGB-D view
frame/time overlay

UTC + monotonic timestamp

C YOLO semantics

boxes + class + confidence
2 Hz inference

UTC + monotonic timestamp

D Environment topology

world DD-GNG nodes/edges
semantic node colours

UTC + monotonic timestamp

E Reachability + path

reachable nodes + selected goal
exact-valid preview

UTC + monotonic timestamp

F Performance

stage latency + CPU/GPU/RAM
trial result and event timeline

UTC + monotonic timestamp

Keep run ID, target object, stage, and frame timestamp visible in every recorded panel.